

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

ME-6004-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Metal Cutting and Machine Tools

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is machine tools? How do you classify the various types of machine tools? 7
b) Draw a labelled diagram of lathe. How machining time can be calculated for turning operation? 7
2. a) What is turret lathe? How it is different from the capstan lathe? 7
b) Explain the tool signature of a SPCT. 7
3. a) What is grinding process? Discuss the advantages of grinding process over the other cutting processes. 7
b) How a universal grinder differs from plain cylindrical grinder? 7
4. a) What is centre less grinding? Explain external centre less grinding operations. 7
b) Explain the terms: Grit, Grade and structure of grinding wheels. 7

ME-6004-CBGS

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[2]

5. a) What do you understand by climb milling and conventional milling? How can you compare both milling processes? 7
b) Draw a labelled diagram of a radial drilling machine and describe its working in brief. 7
6. a) Discuss various cutting parameter in shaper. Derive an expression for calculating the cutting speed. 7
b) How the gears are manufactured by die-casting? Write down its advantages and limitations. 7
7. a) Explain the mechanism of flank wear and crater wear. 7
b) A tool life of 100 min is recorded in a machining test while machining at 25 m/min and 6 min at 70 m/min on mild steel work piece. 7
i) Establish the tool life equation
ii) What is cutting speed for 1 min tool life
8. Write down short notes on any four of the followings: 14
i) Thread production on lathe
ii) Dividing head
iii) Broaching
iv) Surface finish
v) Machinability
